System Identification (Part 1)

Subspace Identification Methods, Multivariate Regression, CVA

Assoc. Prof. Karl Ezra Pilario
Process Systems Engineering Laboratory
Department of Chemical Engineering

© 2023 Karl Ezra S. Pilario, PhD.

Outline

- What is System Identification?
    - Subspace Identification Methods
        - Canonical Variate Analysis (CVA)
- Continuous vs. Discrete SS Models
- Generate simulated data in MATLAB Simulink
- MATLAB System Identification Toolbox

© 2023 Karl Ezra S. Pilario, PhD.

System Identification (SysID)

- Identifying *mathematical models* from the process data generated by a SISO / MIMO system.

- For LTI models, system identification refers to the problem of finding $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$ that best fits the process data.

White-box Model

$$\frac{dh}{dt} = \frac{1}{\rho A}(F_{\text{in}} - F_{\text{out}})$$

$$\frac{dT_{\text{out}}}{dt} = \frac{F_{\text{in}}}{\rho A h}(T_{\text{in}} - T_{\text{out}}) + \frac{Q}{\rho c_p A h}$$

$$F_{\text{out}} = C_v\sqrt{h}$$

- Given the data, fit the values of ρ, A, C_v, c_p.
- Task: Parameter Estimation
- Based on *first-principles* (chemical / physical principles)
- Parameters have meaning.
- Time-consuming to identify.
- Restrictive assumptions.

Black-box Model

$$\frac{d\boldsymbol{x}}{dt} = \dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}$$

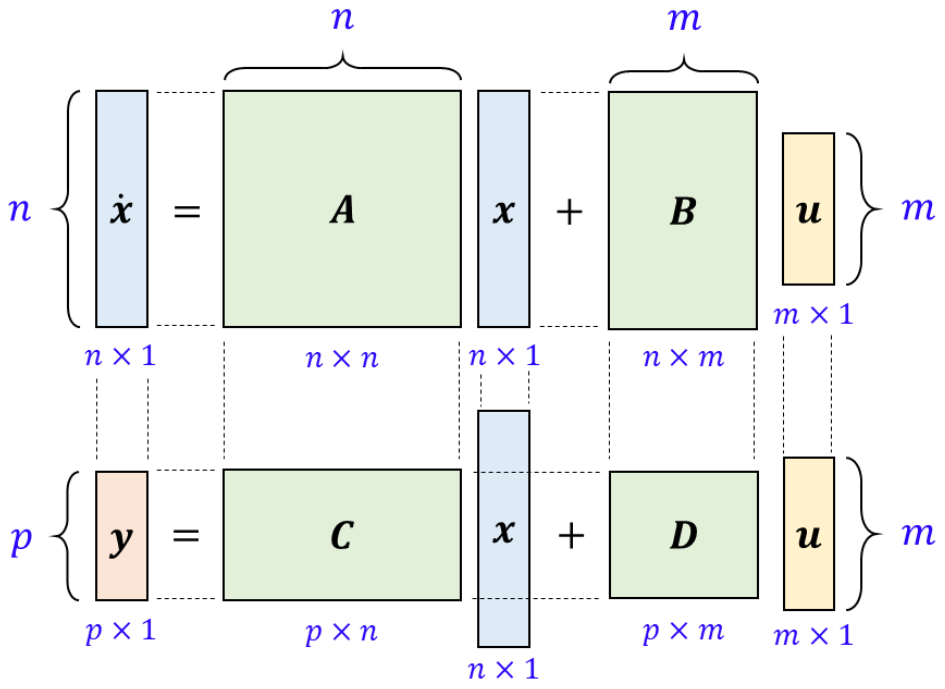
$$\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x} + \boldsymbol{D}\boldsymbol{u}$$

- Given the data, estimate the $\dot{\boldsymbol{x}}$ and fit the values of $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$.
- Task: System Identification
- Based on an *assumed* model structure with right complexity.
- Parameters have no meaning.
- Easier to identify, esp. for MIMO.
- More flexible to many scenarios.

State-space Model

A linear time-invariant (**LTI**) state-space model can be written as:

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}$$
$$\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x} + \boldsymbol{D}\boldsymbol{u}$$

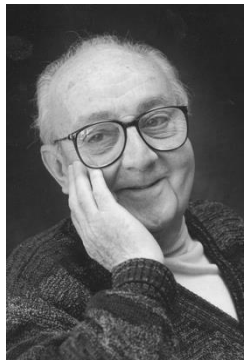


© 2023 Karl Ezra S. Pilario, PhD.

System Identification

Some **limitations** of system identification:

- *"All models are wrong, but some are useful"* – George Box (1976)



For example, the law PV = nRT relating pressure P, volume V and temperature T of an "ideal" gas via a constant R is not exactly true for any real gas, but it frequently provides a useful approximation and furthermore its structure is informative since it springs from a physical view of the behavior of gas molecules.

For such a model there is no need to ask the question "Is the model true?". If "truth" is to be the "whole truth" the answer must be "No". The only question of interest is "Is the model illuminating and useful?".



Occam's Razor

(Law of Parsimony)

- *Among competing models that explain the same behavior, the simplest one is preferred—the one with the fewest parameters.*
- *Among competing theories that explain the same phenomenon, the one making the fewest assumptions is preferred.*

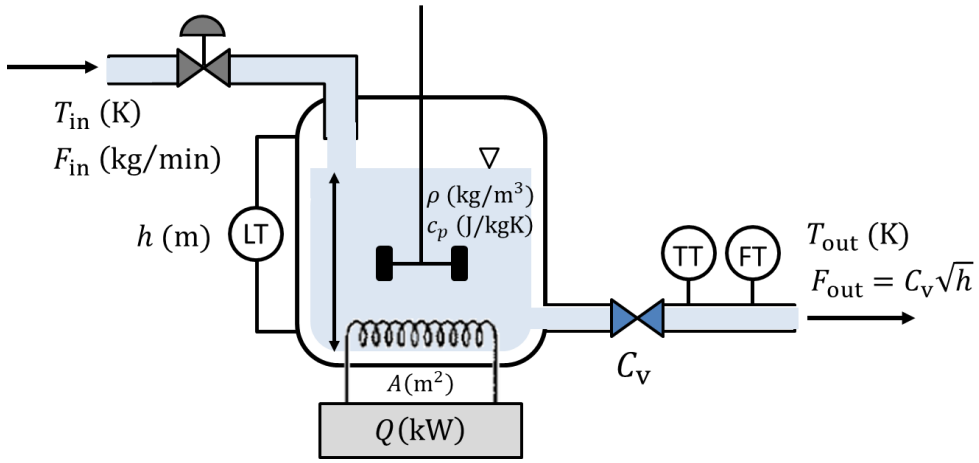
Any model is only a simplification of reality.
Choose the right model complexity for the job!

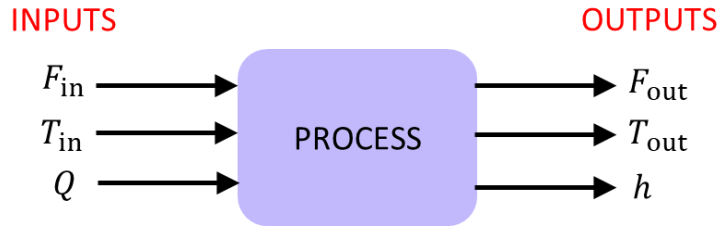
© 2023 Karl Ezra S. Pilario, PhD.

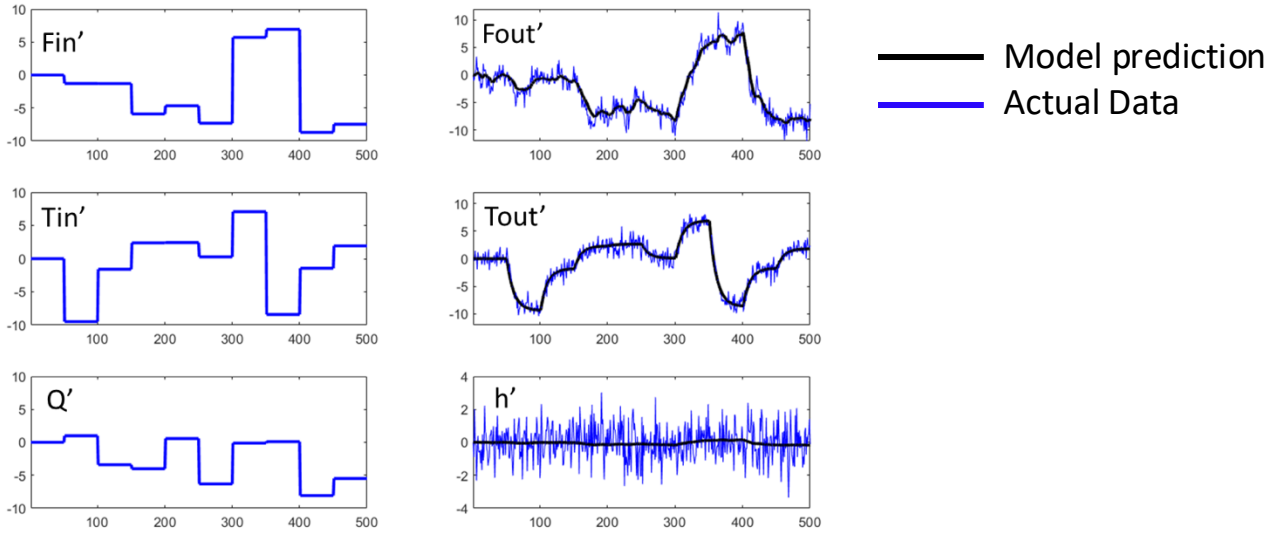
System Identification

Some **limitations** of system identification:

- *"All models are wrong, but some are useful"* – George Box (1976)
- Any model is only a simplification of reality.
  Choose the right model complexity for the job!
- Models are usually "black-box."
  Only input-output behavior is modelled, and anything about the internal is unknown. No meaning is given to the fitted model parameters.
- Measured data may be corrupted by noise.
  We need a mechanism to *filter* the noise, so that only the desired signal is left for system identification.







© 2023 Karl Ezra S. Pilario, PhD.

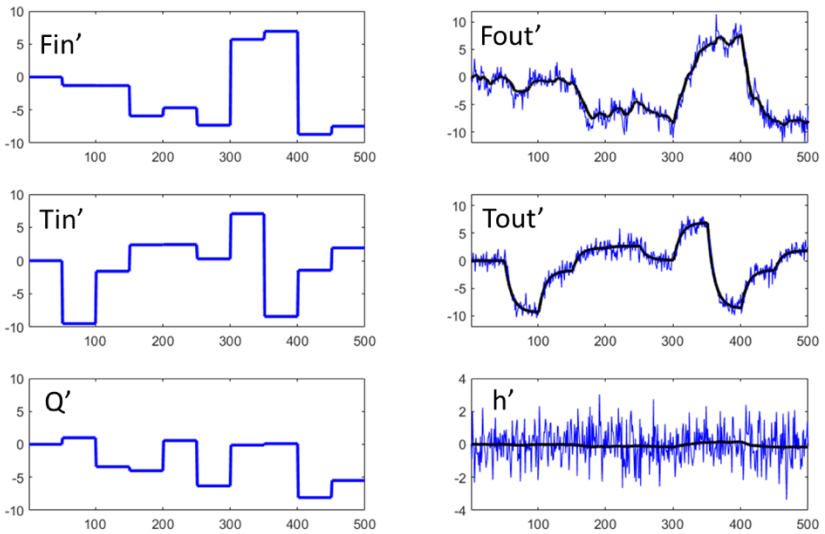
System Identification

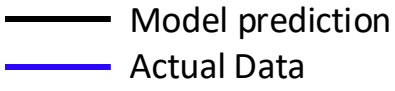
Basic Steps in System Identification

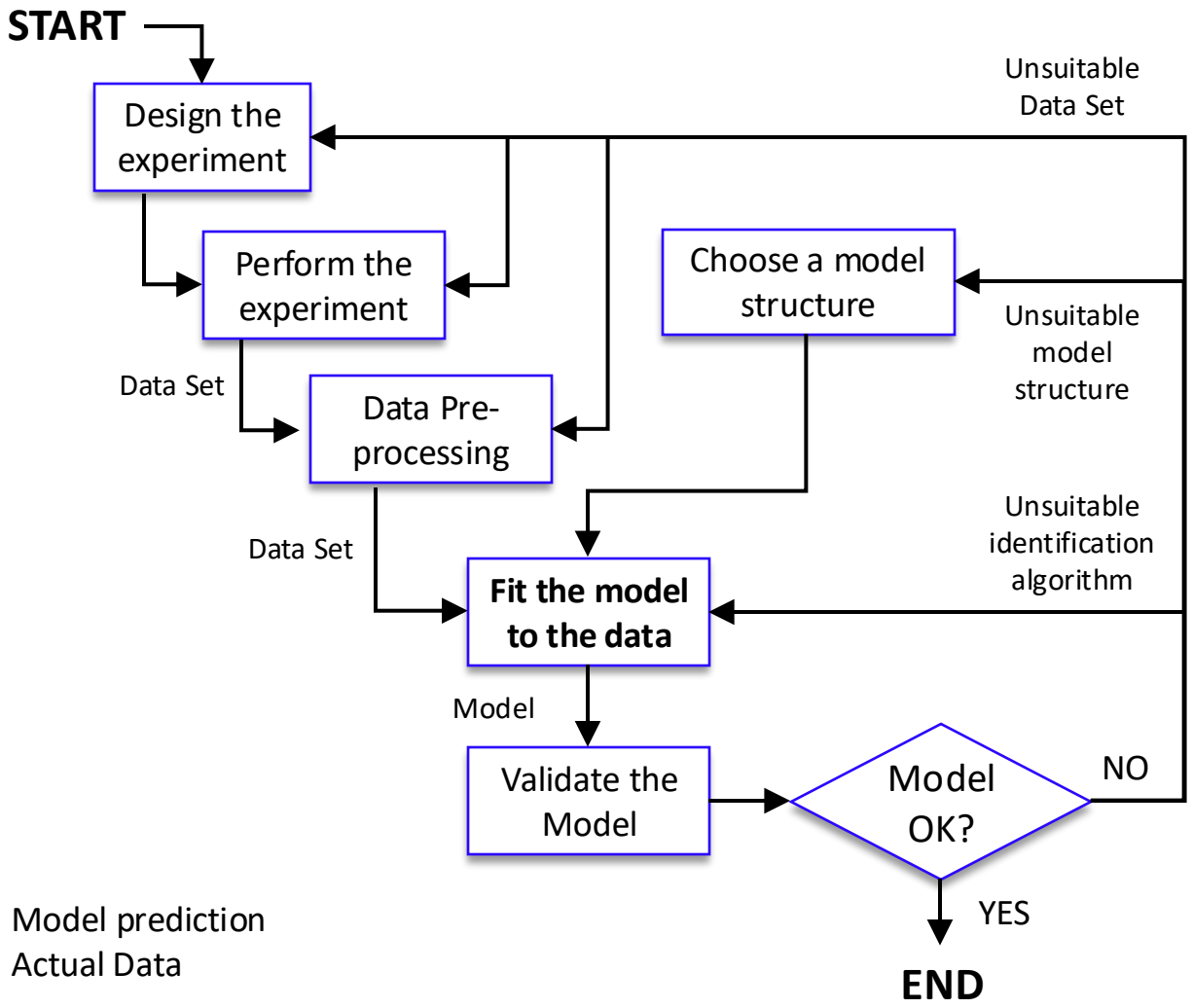
(Verhaegen and Verdult, 2007)

There are 3 ingredients to creating a good model:

1. A **data set** that is rich with information.
2. An appropriate **model structure.**
3. An **algorithm** for fitting the model to data.



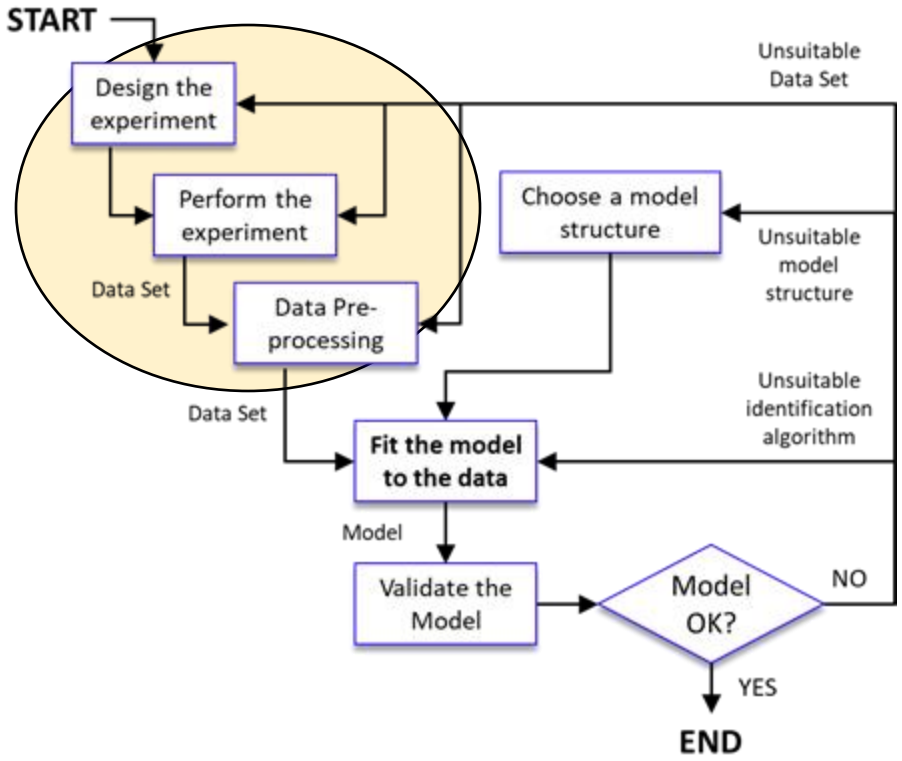




© 2023 Karl Ezra S. Pilario, PhD.

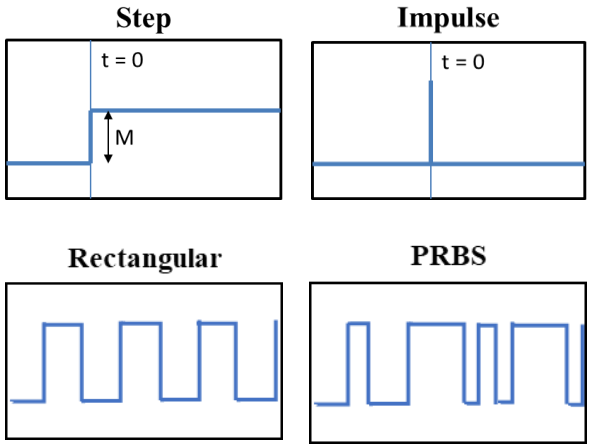
System Identification

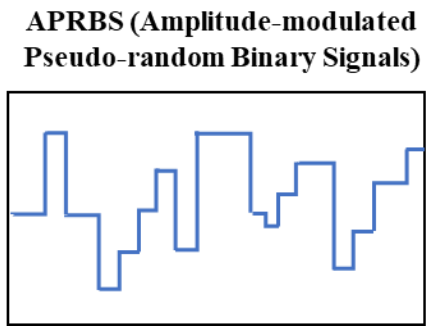
Basic Steps in System Identification
(Verhaegen and Verdult, 2007)



Input Excitations

- The input signal is the only way to *perturb* the process in order to bring out information about its dynamic behavior.
- An example of input change is opening / closing a control valve.
- Designing input signals is not simple. Must think about:
  - time constraints of the experiment
  - safety constraints of the experiment
  - coverage of possible operating conditions
  - persistency and frequency of input excitation
- Typical input signals for system identification:

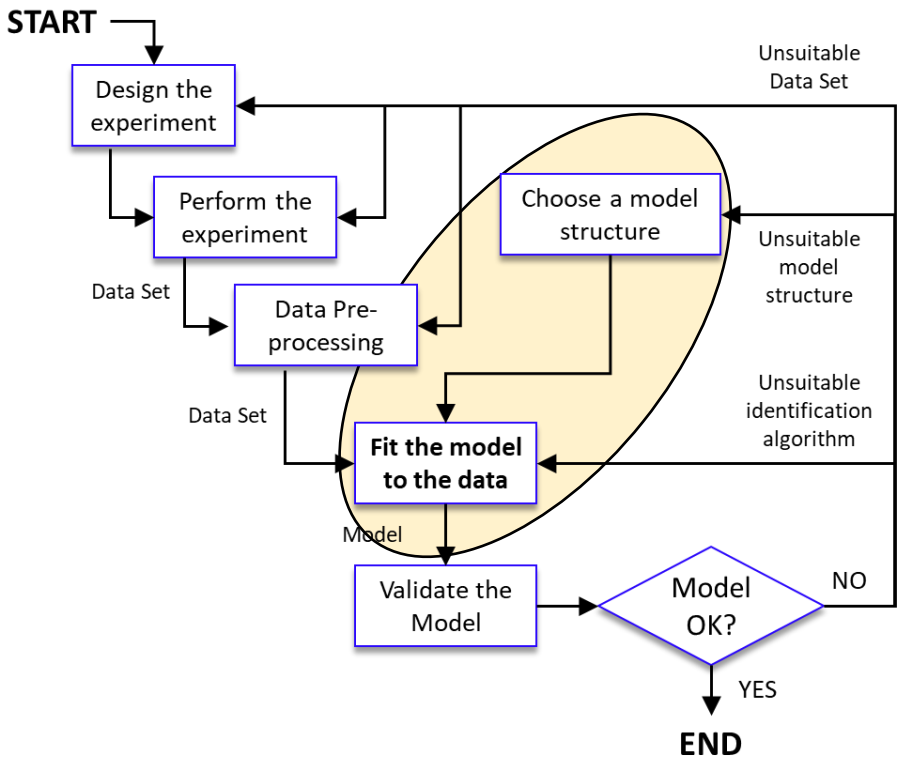




© 2023 Karl Ezra S. Pilario, PhD.

System Identification

Basic Steps in System Identification
(Verhaegen and Verdult, 2007)



"Choosing the model structure is possibly the hardest and most subjectively influenced decision [in system identification]."

– Nelles, 2001

Model structure: State-space Model

Model Fitting:

Subspace Identification Methods (**SIM**)	Prediction Error Methods (**PEM**)

Algorithm:

Training data $\{\boldsymbol{u}_i, \boldsymbol{y}_i\}$

Matrix projections

State sequence, $\boldsymbol{x}_i$

Multivariate Regression

Final Model

Training data $\{\boldsymbol{u}_i, \boldsymbol{y}_i\}$

Initialize model parameters

Model

Iterate

Final Model

© 2023 Karl Ezra S. Pilario, PhD.

System Identification

Advantages of **SIMs**:

- Non-iterative, faster to train
- Optimization is convex
- Suitable for MIMO processes

Disadvantages of **SIMs**:

- Limited to state-space models
- Difficult to apply to nonlinear data

Advantages of **PEMs**:

- Can apply to structures other than state-space models
- Can adapt to nonlinear data

Disadvantages of **PEMs**:

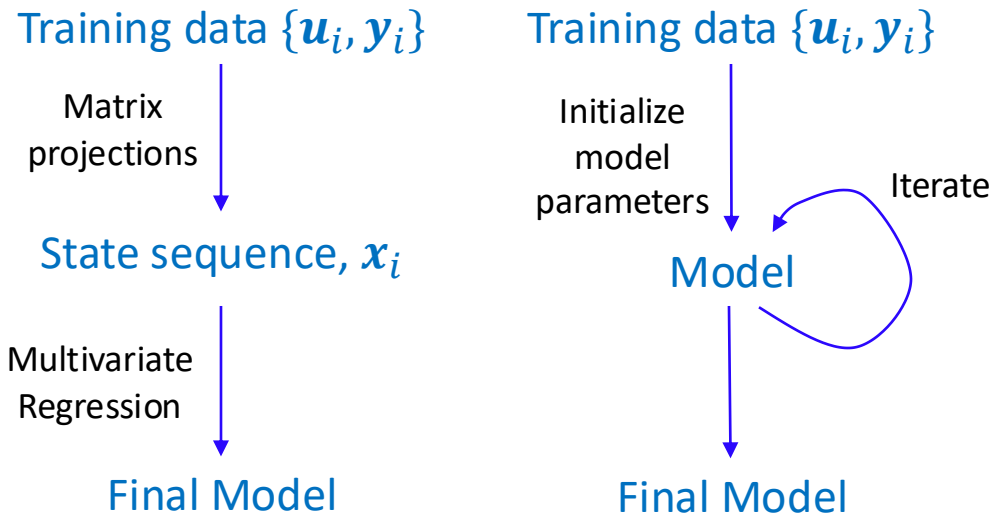
- Requires an iterative solution
- Optimization is non-convex

Model structure: State-space Model

Model Fitting:

Subspace Identification Methods (**SIM**)	Prediction Error Methods (**PEM**)

Algorithm:



© 2023 Karl Ezra S. Pilario, PhD.

System Identification

Subspace Identification Methods (SIM)

- CVA (Canonical Variate Analysis)
  - Larimore (1990), doi: 10.1109/CDC.1990.203665
- MOESP (Multivariable Output-Error State sPace)
  - Verhaegen and Dewilde (1992), doi: 10.1080/00207179208934363
- N4SID (Numerical Algorithms for State-space Subspace System Identification)
  - Van Overschee and De Moor (1993), doi: 10.1016/S1474-6670(17)48221-8
- Unified Theory of Subspace Algorithms
  - Van Overschee and De Moor (1995), doi: 10.1016/0005-1098(95)00072-0

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Prediction Error Methods (PEM)

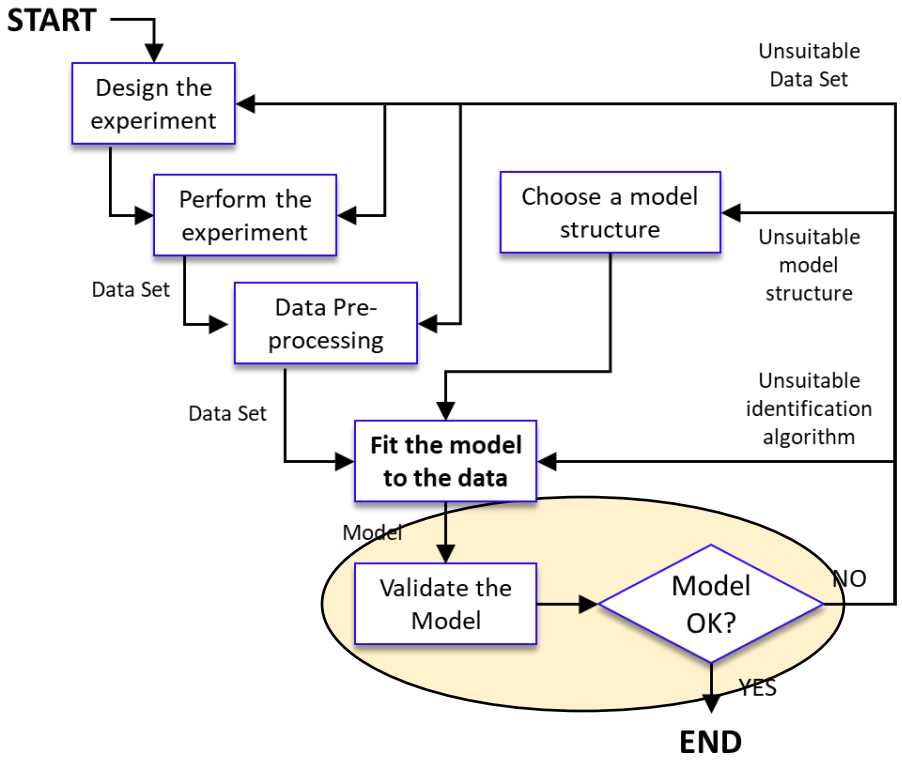
- Polynomial models: $A(q)y(t) = \frac{B(q)}{F(q)}u(t) + \frac{C(q)}{D(q)}e(t)$
  - FIR model
  - AR, ARX, ARMA, ARMAX model
  - Output error (OE) model
  - Box-Jenkins (BJ) model
- Hammerstein-Weiner Models
- Neuro-Fuzzy Models
- Piecewise Affine Models
- Local Linear Model Trees (LOLIMOT)
- Neural Networks
  - Recurrent Neural Nets / LSTMs
  - Multi-layer Feed Forward Networks
  - NARX (Nonlinear Autoregressive with eXogenous Inputs)
  - Radial Basis Function Networks
- Least-squares Support Vector Machines

$$\dot{x} = f(x, u, t)$$
$$y = f(x, u, t)$$

© 2023 Karl Ezra S. Pilario, PhD.

System Identification

Basic Steps in System Identification
(Verhaegen and Verdult, 2007)



How to evaluate a model's goodness of fit?

1. Mean Squared Error (**MSE**)

$$MSE = \frac{1}{N}\sum_i \left(y_i^{\text{ref}} - y_i^{\text{pred}}\right)^2$$

MSE = 0 (perfect fit)
MSE = Inf (worst fit)

2. Normalized Mean Squared Error (**NMSE**)

$$NMSE = \frac{\sum_i \left(y_i^{\text{ref}} - y_i^{\text{pred}}\right)^2}{\sum_i \left(y_i^{\text{ref}} - \text{mean}(\boldsymbol{y}^{\text{ref}})\right)^2}$$

NMSE = 0 (perfect fit)
NMSE = Inf (worst fit)

3. Coefficient of Determination ($\boldsymbol{R^2}$)

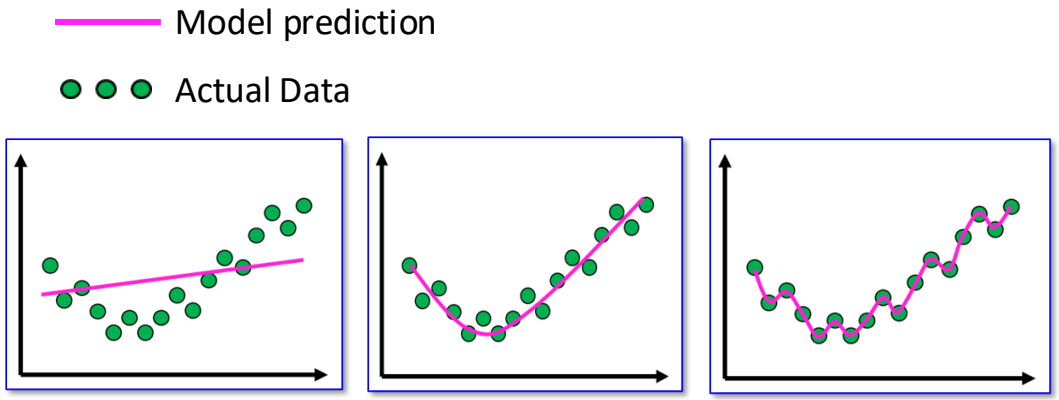
$$R^2 = 1 - NMSE$$

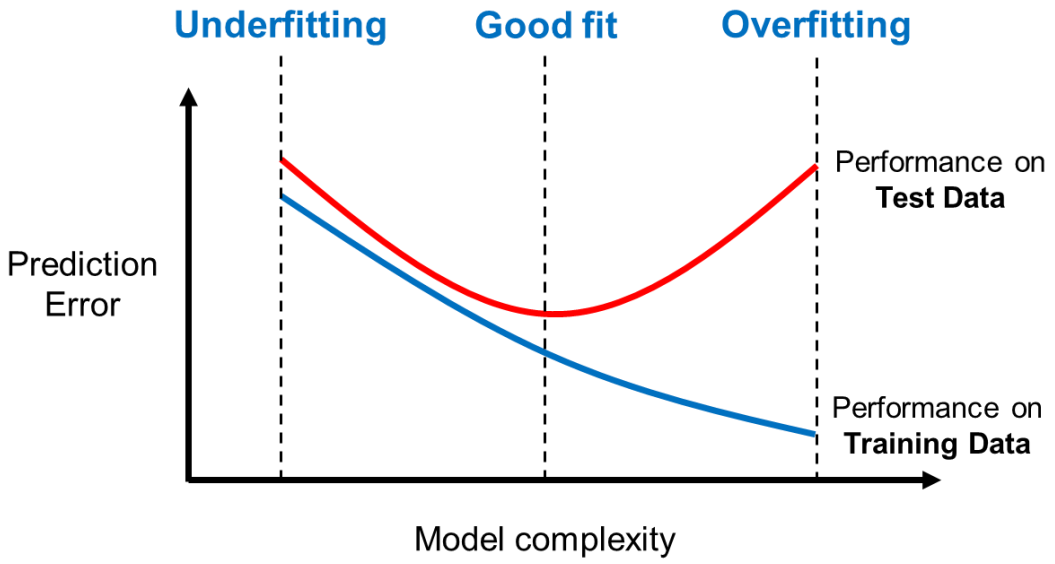
R^2 = 1 (perfect fit)
R^2 = -Inf (worst fit)

Note: If $R^2 = 0$, then our model is no better than always reporting the *mean* of y^{ref} as the prediction y^{pred}.

© 2023 Karl Ezra S. Pilario, PhD.

System Identification





How to evaluate a model's goodness of fit?

1. Mean Squared Error (**MSE**)

$$MSE = \frac{1}{N}\sum_i \left(y_i^{\text{ref}} - y_i^{\text{pred}}\right)^2$$

2. Normalized Mean Squared Error (**NMSE**)

$$NMSE = \frac{\sum_i \left(y_i^{\text{ref}} - y_i^{\text{pred}}\right)^2}{\sum_i \left(y_i^{\text{ref}} - \text{mean}(\boldsymbol{y}^{\text{ref}})\right)^2}$$

3. Coefficient of Determination ($\boldsymbol{R^2}$)

$$R^2 = 1 - NMSE$$

- **In practice, these metrics are not enough!**
- Must test if the model serves its purpose.
- You can also check for the autocorrelation of the error. If the error has no correlation, then the model captured all possible correlations.
- Think about **Occam's Razor** (Law of Parsimony). Simple models must be tried first!
- Must avoid overfitting and underfitting by adjusting the model complexity (e.g. model order / no. of parameters)

© 2023 Karl Ezra S. Pilario, PhD.

Outline

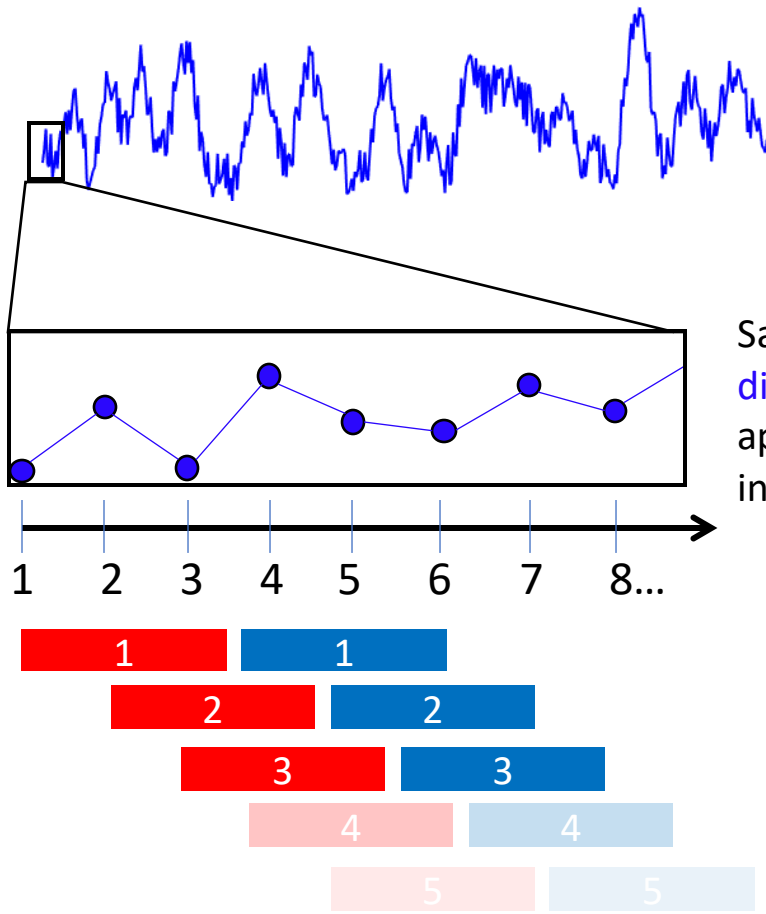
- What is System Identification?
    - Subspace Identification Methods
        - **Canonical Variate Analysis (CVA)**
- Continuous vs. Discrete SS Models
- Generate simulated data in MATLAB Simulink
- MATLAB System Identification Toolbox

© 2023 Karl Ezra S. Pilario, PhD.

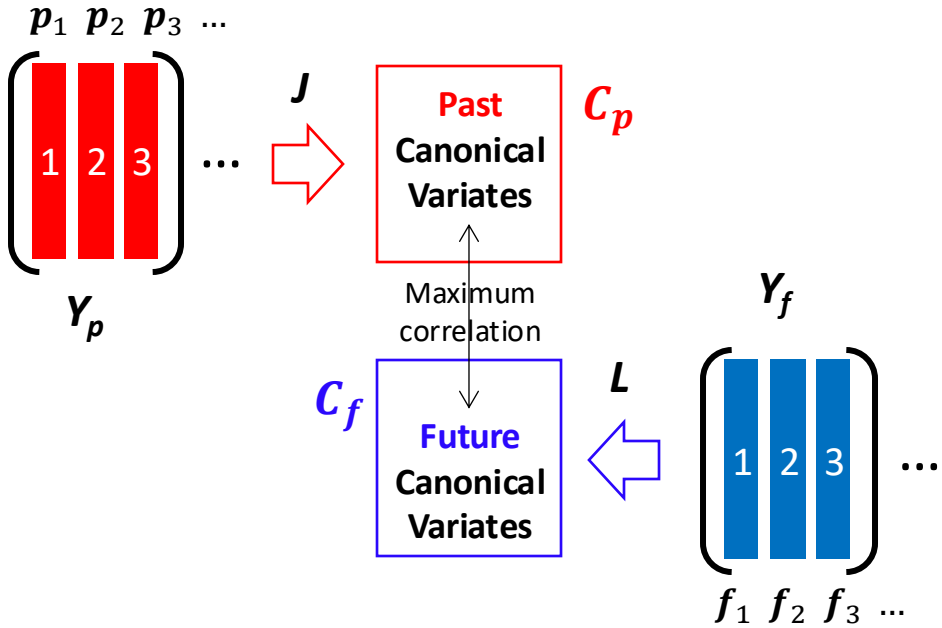
Canonical Variate Analysis

Training a model from maximizing the *correlation* between **past** data and **future** data.

Process historical data:



Samples are discrete; they only appear at regular instants in time.



$$C_p = JY_p$$

$$C_f = LY_f$$

The goal of **CVA** is to find projection matrices J and L that maximizes the correlation between C_p and C_f.

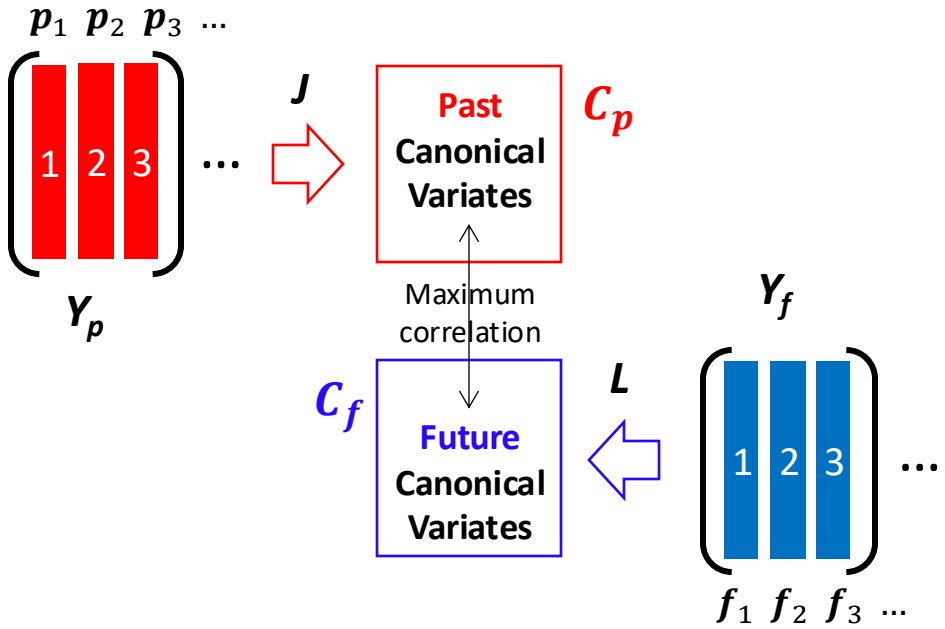
© 2023 Karl Ezra S. Pilario, PhD.

Canonical Variate Analysis

Think of each *row vector* in the projection matrices as weights:

$$C_f = \begin{bmatrix} a_1^T \\ a_2^T \\ a_3^T \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & \cdots \end{bmatrix} \begin{matrix} \\ \\ \\ \end{matrix}$$
$$\begin{matrix} f_1 & f_2 & f_3 & \cdots \end{matrix}$$

$$C_p = \begin{bmatrix} b_1^T \\ b_2^T \\ b_3^T \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & \cdots \end{bmatrix} \begin{matrix} \\ \\ \\ \end{matrix}$$
$$\begin{matrix} p_1 & p_2 & p_3 & \cdots \end{matrix}$$



The correlation between C_p and C_f is given by:

$$\rho_i = \frac{(a_i^T f_i)(b_i^T p_i)^T}{\sqrt{(a_i^T f_i)(a_i^T f_i)^T}\sqrt{(b_i^T p_i)(b_i^T p_i)^T}}$$

$$\rho_i = \frac{a_i^T f_i p_i^T b_i}{\sqrt{a_i^T f_i f_i^T a_i}\sqrt{b_i^T p_i p_i^T b_i}}$$

From statistics:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$C_p = JY_p$$
$$C_f = LY_f$$

The goal of **CVA** is to find projection matrices J and L that maximizes the correlation between C_p and C_f.

© 2023 Karl Ezra S. Pilario, PhD.

Canonical Variate Analysis

Think of each *row vector* in the projection matrices as weights:

$$\boldsymbol{C_f} = \begin{bmatrix} a_1^T \\ a_2^T \\ a_3^T \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & \cdots \end{bmatrix}$$

$$\boldsymbol{f}_1 \ \boldsymbol{f}_2 \ \boldsymbol{f}_3 \ \cdots$$

$$\boldsymbol{C_p} = \begin{bmatrix} b_1^T \\ b_2^T \\ b_3^T \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & \cdots \end{bmatrix}$$

$$\boldsymbol{p}_1 \ \boldsymbol{p}_2 \ \boldsymbol{p}_3 \ \cdots$$

The correlation between $\boldsymbol{C_p}$ and $\boldsymbol{C_f}$ is given by:

$$\rho_i = \frac{(\boldsymbol{a}_i^T \boldsymbol{f}_i)(\boldsymbol{b}_i^T \boldsymbol{p}_i)^T}{\sqrt{(\boldsymbol{a}_i^T \boldsymbol{f}_i)(\boldsymbol{a}_i^T \boldsymbol{f}_i)^T}\sqrt{(\boldsymbol{b}_i^T \boldsymbol{p}_i)(\boldsymbol{b}_i^T \boldsymbol{p}_i)^T}}$$

$$\rho_i = \frac{\boldsymbol{a}_i^T \boldsymbol{f}_i \boldsymbol{p}_i^T \boldsymbol{b}_i}{\sqrt{\boldsymbol{a}_i^T \boldsymbol{f}_i \boldsymbol{f}_i^T \boldsymbol{a}_i}\sqrt{\boldsymbol{b}_i^T \boldsymbol{p}_i \boldsymbol{p}_i^T \boldsymbol{b}_i}}$$

To collate all results for $i = 1, 2, \ldots, N$:

$$\rho = \frac{\boldsymbol{a}^T \boldsymbol{\Sigma}_{fp} \boldsymbol{b}^T}{\sqrt{\boldsymbol{a}^T \boldsymbol{\Sigma}_{ff} \boldsymbol{a}}\sqrt{\boldsymbol{b}^T \boldsymbol{\Sigma}_{pp} \boldsymbol{b}}}$$

$$\boldsymbol{\Sigma}_{pp} = \frac{1}{M-1} \boldsymbol{Y}_p \boldsymbol{Y}_p^T$$

$$\boldsymbol{\Sigma}_{ff} = \frac{1}{M-1} \boldsymbol{Y}_f \boldsymbol{Y}_f^T$$

$$\boldsymbol{\Sigma}_{fp} = \frac{1}{M-1} \boldsymbol{Y}_f \boldsymbol{Y}_p^T$$

Let's denote $\boldsymbol{u} = \boldsymbol{\Sigma}_{ff}^{1/2} \boldsymbol{a}$ and $\boldsymbol{v} = \boldsymbol{\Sigma}_{pp}^{1/2} \boldsymbol{b}$.

We can form the maximization problem as:

$$\max_{\boldsymbol{u},\boldsymbol{v}} \quad \rho = \frac{\boldsymbol{u}^T \left(\boldsymbol{\Sigma}_{ff}^{-1/2} \boldsymbol{\Sigma}_{fp} \boldsymbol{\Sigma}_{pp}^{-1/2} \right) \boldsymbol{v}}{(\boldsymbol{u}^T \boldsymbol{u})(\boldsymbol{v}^T \boldsymbol{v})}$$

This part **differs** among CVA, N4SID and MOESP.

If $\boldsymbol{H} = \boldsymbol{\Sigma}_{ff}^{-1/2} \boldsymbol{\Sigma}_{fp} \boldsymbol{\Sigma}_{pp}^{-1/2}$, it can be derived that the maximum solutions are the *singular values* of $\boldsymbol{H}$:

$$\boldsymbol{H} = \boldsymbol{\mathcal{U}} \boldsymbol{S} \boldsymbol{\mathcal{V}}^T$$

Singular Value Decomposition

© 2023 Karl Ezra S. Pilario, PhD.

Canonical Variate Analysis

If $\boldsymbol{H = \Sigma_{ff}^{-1/2}\Sigma_{fp}\Sigma_{pp}^{-1/2}}$, it can be derived that the maximum solutions are the *singular values* of $\boldsymbol{H}$:

$$\boldsymbol{H = \mathcal{U}S\mathcal{V}^T}$$

Singular Value Decomposition

$$\boldsymbol{H} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 \end{pmatrix} \begin{pmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \sigma_3 \end{pmatrix} \begin{pmatrix} \boldsymbol{v}_1^T \\ \boldsymbol{v}_2^T \\ \boldsymbol{v}_3^T \end{pmatrix}$$

$$\boldsymbol{\mathcal{U}} \times \boldsymbol{S} \times \boldsymbol{\mathcal{V}^T}$$

Based on the definitions $\boldsymbol{u = \Sigma_{ff}^{1/2} a}$ and $\boldsymbol{v = \Sigma_{pp}^{1/2} b}$:

$$\boldsymbol{J = \mathcal{V}^T\Sigma_{pp}^{-1/2}} = \begin{pmatrix} b_1^T \\ b_2^T \\ b_3^T \end{pmatrix}$$

$$\boldsymbol{L = \mathcal{U}^T\Sigma_{ff}^{-1/2}} = \begin{pmatrix} a_1^T \\ a_2^T \\ a_3^T \end{pmatrix}$$

$$\boldsymbol{C_p = JY_p}$$

$$\boldsymbol{C_f = LY_f}$$

Important properties of the **SVD** result:

$$\frac{1}{N-1}\boldsymbol{C_p C_p^T} = \frac{1}{N-1}\boldsymbol{J\Sigma_{pp}J^T} = \boldsymbol{I}$$

The covariance of **past** canonical variates is the *identity* matrix.

$$\frac{1}{N-1}\boldsymbol{C_f C_f^T} = \frac{1}{N-1}\boldsymbol{L\Sigma_{ff}L^T} = \boldsymbol{I}$$

The covariance of **future** canonical variates is the *identity* matrix.

$$\frac{1}{N-1}\boldsymbol{C_f C_p^T} = \frac{1}{N-1}\boldsymbol{L\Sigma_{fp}J^T} = \boldsymbol{S}$$

The cross-covariance of **past** and **future** canonical variates is the *diagonal matrix of singular values.*

From here, we take only the first few **past** canonical variates as the states, $\boldsymbol{x}$:

$$\boldsymbol{X = [x_i] = C_p}$$

Finally, we use *multivariate linear regression* to get $\boldsymbol{A, B, C, D}$:

$$\boldsymbol{[\hat{A} \quad \hat{B}] = X_{t+1}\begin{pmatrix} X_t \\ U_t \end{pmatrix}^+}$$

Inputs: $\boldsymbol{U = [u_i]}$

$$\boldsymbol{[\hat{C} \quad \hat{D}] = Y_t\begin{pmatrix} X_t \\ U_t \end{pmatrix}^+}$$

Outputs: $\boldsymbol{Y = [y_i]}$

© 2023 Karl Ezra S. Pilario, PhD.

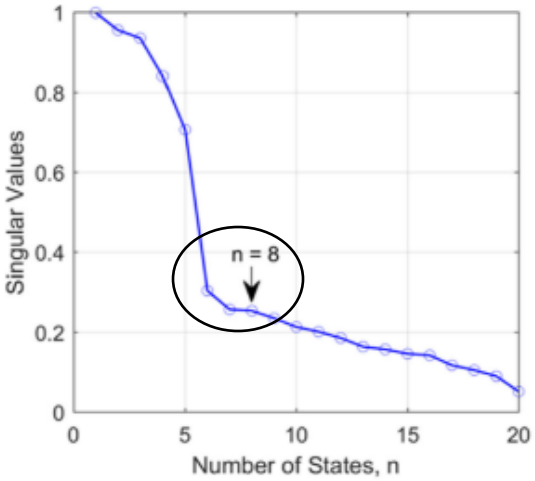
Canonical Variate Analysis

From here, we take only the first few **past** canonical variates as the states, $\boldsymbol{x}$:

$$\boldsymbol{X} = [\boldsymbol{x}_i] = \boldsymbol{C}_p$$

Note: The number (n) of states chosen from the full $\boldsymbol{C}_p$ is called the *model order*.

The value of n can be chosen by finding the **knee** in the plot of singular values ($\sigma_1, \sigma_2, \ldots$).



Finally, we use *multivariate linear regression* to get $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$:

$$[\widehat{\boldsymbol{A}} \quad \widehat{\boldsymbol{B}}] = \boldsymbol{X}_{t+1} \begin{pmatrix} \boldsymbol{X}_t \\ \boldsymbol{U}_t \end{pmatrix}^+$$

$$[\widehat{\boldsymbol{C}} \quad \widehat{\boldsymbol{D}}] = \boldsymbol{Y}_t \begin{pmatrix} \boldsymbol{X}_t \\ \boldsymbol{U}_t \end{pmatrix}^+$$

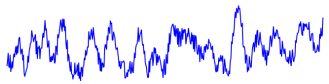
Inputs: $\boldsymbol{U} = [\boldsymbol{u}_i]$

Outputs: $\boldsymbol{Y} = [\boldsymbol{y}_i]$

The *pseudo-inverse*, A^+, is computed as: $A^+ = (A^T A)^{-1} A^T$

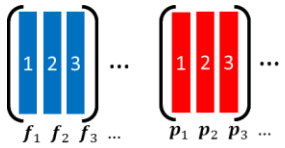
The lower the n, the simpler the model. However, too low n leads to underfitting. Too high n leads to overfitting.

Summary: Canonical Variate Analysis

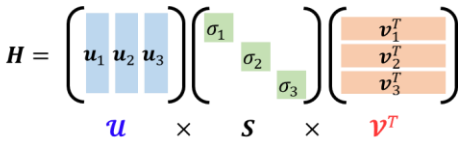


Input: $\boldsymbol{U}$
Output: $\boldsymbol{Y}$

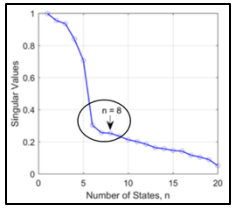
Form **past** and **future** matrices:



Solve the SVD of $\boldsymbol{H}$:



Choose the *model order*:



Solve for **A, B, C, D**:

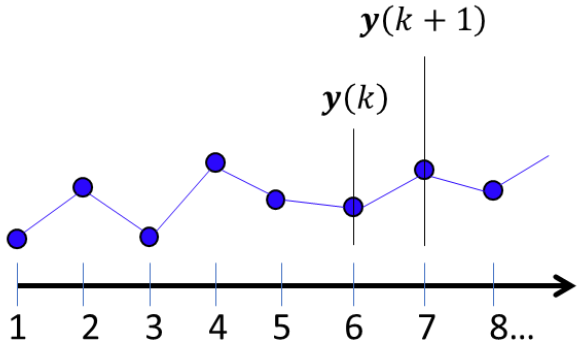
$$[\widehat{\boldsymbol{A}} \quad \widehat{\boldsymbol{B}}] = \boldsymbol{X}_{t+1} \begin{pmatrix} \boldsymbol{X}_t \\ \boldsymbol{U}_t \end{pmatrix}^+$$

$$[\widehat{\boldsymbol{C}} \quad \widehat{\boldsymbol{D}}] = \boldsymbol{Y}_t \begin{pmatrix} \boldsymbol{X}_t \\ \boldsymbol{U}_t \end{pmatrix}^+$$

© 2023 Karl Ezra S. Pilario, PhD.

Continuous vs. Discrete SS models

- The output of typical system identification methods are actually discrete models:



Discrete state-space model
$(k = 1, 2, 3, \dots)$

$$\boldsymbol{x}(k+1) = \boldsymbol{A}\,\boldsymbol{x}(k) + \boldsymbol{B}\,\boldsymbol{u}(k)$$
$$\boldsymbol{y}(k) = \boldsymbol{C}\,\boldsymbol{x}(k) + \boldsymbol{D}\,\boldsymbol{u}(k)$$

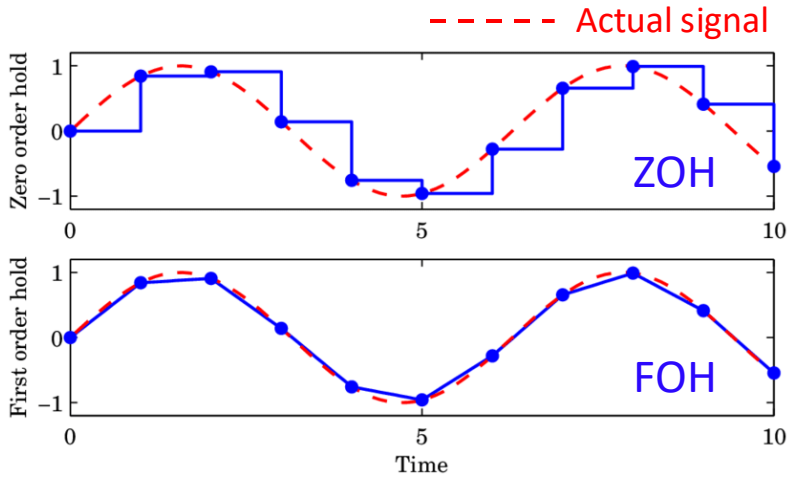
Or sometimes…

$$\boldsymbol{x}_{k+1} = \boldsymbol{A}\,\boldsymbol{x}_k + \boldsymbol{B}\,\boldsymbol{u}_k$$
$$\boldsymbol{y}_k = \boldsymbol{C}\,\boldsymbol{x}_k + \boldsymbol{D}\,\boldsymbol{u}_k$$

Continuous state-space model
$(t \in \Re^+)$

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\,\boldsymbol{x}(t) + \boldsymbol{B}\,\boldsymbol{u}(t)$$
$$\boldsymbol{y}(t) = \boldsymbol{C}\,\boldsymbol{x}(t) + \boldsymbol{D}\,\boldsymbol{u}(t)$$

For discrete-time models, zero-order hold (ZOH) is commonly assumed.



Connection between discrete **ABCD** (zero-order hold) and continuous **ABCD**:

$$\boldsymbol{A}_d = e^{\boldsymbol{A}_c k} \qquad \boldsymbol{C}_d = \boldsymbol{C}_c$$

$$\boldsymbol{B}_d = \int_0^k e^{\boldsymbol{A}_c \lambda}\, d\lambda\ \boldsymbol{B}_c \qquad \boldsymbol{D}_d = \boldsymbol{D}_c$$

Where $e^{\boldsymbol{A}}$ is the matrix exponential.

In MATLAB, we can use `c2d` and `d2c` built-in functions to convert from continuous to discrete and vice versa.

© 2023 Karl Ezra S. Pilario, PhD.

Outline

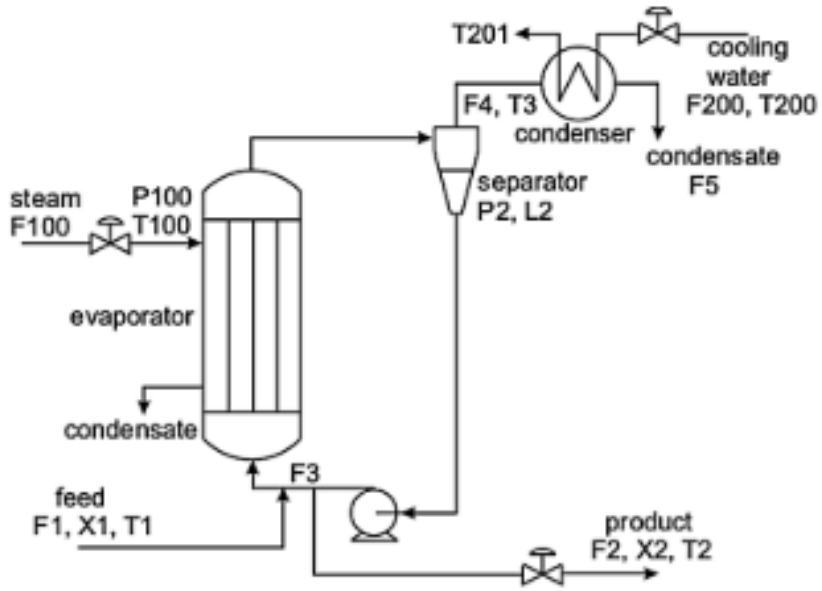
- What is System Identification?
    - Subspace Identification Methods
        - Canonical Variate Analysis (CVA)
- Continuous vs. Discrete SS Models
- **Generate simulated data in MATLAB Simulink**
- **MATLAB System Identification Toolbox**

© 2023 Karl Ezra S. Pilario, PhD.

Canonical Variate Analysis

Example:

Newell-Lee Evaporator System (Newell, Lee, 1989)



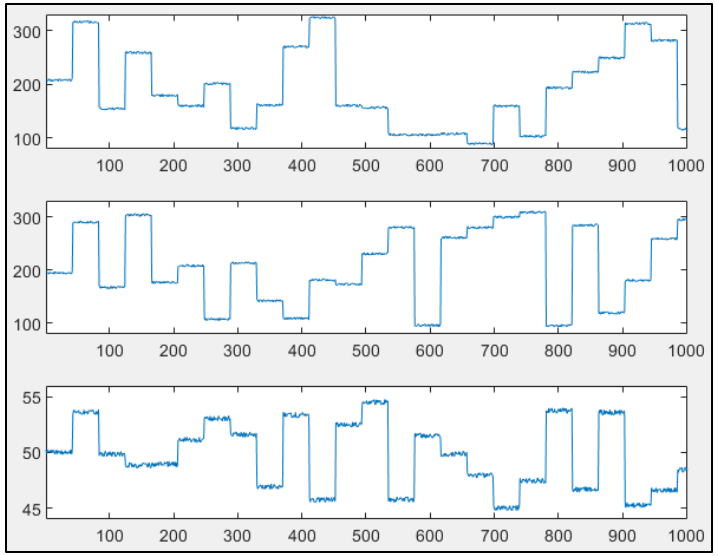
- Boils away the liquor from feed to produce a concentrate using a forced-circulation evaporator.
- Uses steam and cooling water utilities.
- There is one control loop for controlling the level in the separator by manipulating the product valve.
- Highly nonlinear system.

Inputs:

F200

P100

F3

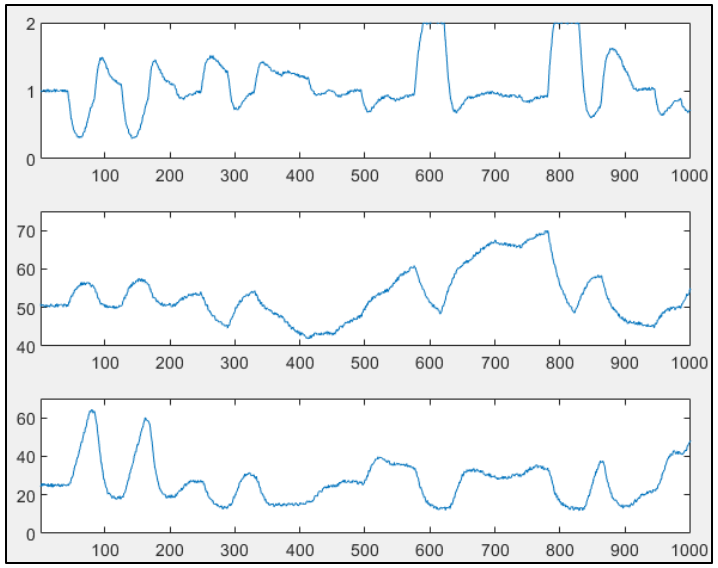


Outputs:

L2

P2

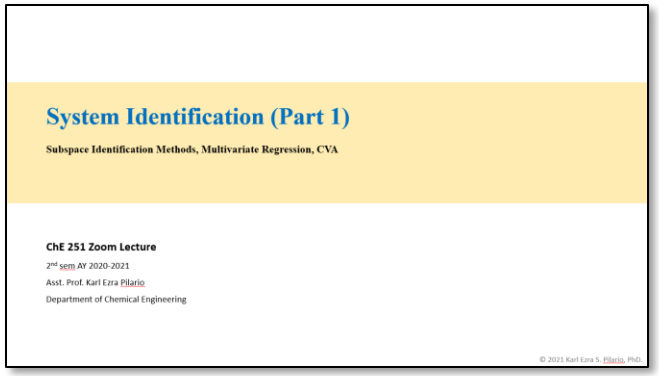
X2



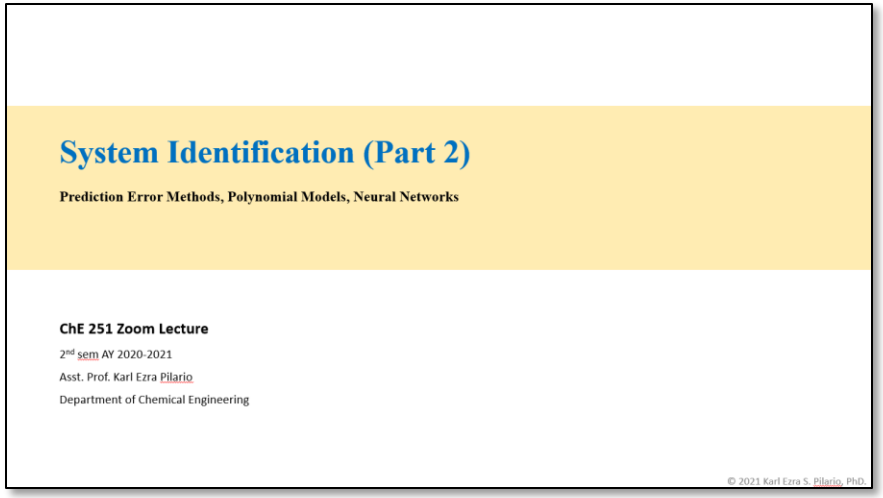
© 2023 Karl Ezra S. Pilario, PhD.

Outline

- System Identification
- Subspace Identification Methods
  - Canonical Variate Analysis (CVA)
- Continuous vs. Discrete SS Models
- Generate simulated data in MATLAB Simulink
- MATLAB System Identification Toolbox



Next week:



© 2023 Karl Ezra S. Pilario, PhD.